

| Capacitors                                |     |                 |         |               |                      |
|-------------------------------------------|-----|-----------------|---------|---------------|----------------------|
| Designators                               | Qty | Value           | Package | Tol           | Voltage              |
| C1,C2,C3,C9,C13,C23,C26                   | 15  | 10 $\mu$ F      | 805     | $\pm 10-20\%$ | $\geq 20$ V, X7R/X5R |
| C4,C8,C14,C15,C16,C17,C19,C20,C22,C24,C25 | 25  | 0.1 $\mu$ F     | 603     | $\pm 10\%$    | $\geq 10$ V, X7R     |
| C5,C21                                    | 5   | 1 $\mu$ F       | 805     | $\pm 10\%$    | $\geq 10$ V, X7R     |
| C6,C7                                     | 5   | 4.7 $\mu$ F     | 805     | $\pm 10-20\%$ | $\geq 10$ V, X7R     |
| C10,C11                                   | 8   | 22 $\mu$ F      | 805     | $\pm 20\%$    | $\geq 20$ V, X7R     |
| C18                                       | 5   | 2.2 $\mu$ F     | 805     | $\pm 10\%$    | $\geq 10$ V, X7R     |
| CY1,CY2                                   | 5   | 12 pF           | 603     | $\pm 5\%$     | $\geq 50$ V, C0G/NP0 |
| Resistors                                 |     |                 |         |               |                      |
| Designators                               | Qty | Value           | Package | Tol           | Power                |
| R1,R4,R9,R14,R16,R17,R18,R24,R15,R19,R20  | 25  | 10 k $\Omega$   | 603     | $\pm 1\%$     | 1/10 W               |
| R2                                        | 5   | 3.1k $\Omega$   | 603     | $\pm 1\%$     | 1/10 W               |
| R5                                        | 5   | 3.56 k $\Omega$ | 603     | $\pm 1\%$     | 1/10 W               |
| R7,R8                                     | 5   | 5.1 k $\Omega$  | 603     | $\pm 1\%$     | 1/10 W               |
| R10,R11                                   | 5   | 6 k $\Omega$    | 603     | $\pm 1\%$     | 1/10 W               |
| R12                                       | 5   | 470 k $\Omega$  | 805     | $\pm 1\%$     | 1/8 W                |
| R13                                       | 5   | 150 k $\Omega$  | 805     | $\pm 1\%$     | 1/8 W                |
| R21,22,23                                 | 8   | 2.2 k $\Omega$  | 603     | $\pm 1\%$     | 1/10 W               |
|                                           | 5   | 1.50 k $\Omega$ | 603     | $\pm 1\%$     | 1/10 W               |
|                                           | 5   | 1.8k $\Omega$   | 603     | $\pm 1\%$     | 1/10 W               |
| DIODE                                     |     |                 |         |               |                      |
| Designators                               | Qty | Part / Value    | Package | Note          |                      |
| D1                                        | 3   | ESD9B5.0 ST5G   | SOD-923 |               |                      |

|          |   |                  |         |  |  |
|----------|---|------------------|---------|--|--|
|          | 3 | ESD9X5.0<br>ST5G | SOD-923 |  |  |
| D2,D4,D6 | 8 | LED<br>Yellow    | 603     |  |  |
| D3,D5    | 5 | LED Red          | 603     |  |  |

| IC & Modules          |     |                                 |                |                                            |
|-----------------------|-----|---------------------------------|----------------|--------------------------------------------|
| Designators           | Qty | Part                            | Package        | Note                                       |
| U1                    | 2   | BQ24075RGT                      | VQFN-16 3×3 mm |                                            |
| U2                    | 2   | TPS63070RNMR                    | VQFN           |                                            |
| U3                    | 2   | USBLC6-2SC6                     | SOT-23-6       |                                            |
| U4                    | 2   | STM32F411CEU6                   | UFQFPN-48      |                                            |
| U5                    | 2   | MDBT42Q-512K                    | Module         |                                            |
| DWM1                  | 2   | DWM1000                         | Module         |                                            |
| Y1                    | 3   | NX3225SA-24<br>MHz 8pF          | 3.2×2.5-4pin   |                                            |
| Magnetics / Bead      |     |                                 |                |                                            |
| Designators           | Qty | Part / Value                    | Package        | Note                                       |
| L1                    | 3   | 1.5 $\mu$ H                     | 1008 (2520)    | $I_{sat} \geq 1.5$ A,<br>DCR thấp          |
| FB1                   | 3   | Ferrite bead                    | 603            | 600 $\Omega$ at 100<br>MHz, $I \geq 0.5$ A |
| Connectors & Switches |     |                                 |                |                                            |
| Designators           | Qty | Part                            | Package        | Note                                       |
| J2                    | 3   | USB-C 16P HRO<br>TYPE-C-31-M-12 | Receptacle     | <a href="#">Link</a>                       |
| J1                    | 3   | BM02B-SRSS-TB                   | JST-SH 2-pin   |                                            |
| J3                    | 3   | BM04B-SRSS-TB                   | JST-SH 4-pin   |                                            |
| J4                    | 3   | BM05B-SRSS-TB                   | JST-SH 5-pin   |                                            |
| SW1                   | 3   | DIP-3, 1.27 mm                  | SMD            |                                            |
| SW2                   | 5   | SW Tact 2P<br>3x4mm             | SMD            | <a href="#">Link</a>                       |
| SW3                   | 3   | SPDT PCM12                      | SMD            | <a href="#">Link</a>                       |
| F1                    | 3   | Polyfuse 0.5A                   | 1206           |                                            |
| Other                 |     |                                 |                |                                            |

| Designators | Qty | Part                           | Package | Note                 |
|-------------|-----|--------------------------------|---------|----------------------|
|             | 1   | Pin Li-Polymer<br>3.7V 1000mAh | 503450  | <a href="#">Link</a> |